MSE 310 Project Report
Automated Widget Assembly

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Executive Summary

This report outlines the efforts of Group 6 to understand and enhance the functionality of the ICT3 system. Our main objective was to develop an automated solution that optimizes the process of sorting, assembling, inspecting, and classifying components—specifically metal pegs and plastic rings. We achieved this by creating an advanced algorithm, programmed in LabView, which manages the system's sensors and actuators. The report delves into the technical details of the ICT3 system, providing an overview of its components, the algorithm's structure, and the flowcharts used to guide its operation. Additionally, we highlight key challenges encountered during the project and suggest potential improvements for future iterations of the system.

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1. Introduction

Industrial automation is a key driver of productivity, accuracy, and safety in modern manufacturing. This project aimed to automate and optimize the operation of a conveyor system used in the assembly of metal pegs and plastic rings. The goal was to create an efficient process that can automatically sort, assemble, inspect, and classify the components, while minimizing human intervention. Using the ICT3 system, our team integrated sensors, actuators, and LabView software to achieve this automation. This report provides an in-depth look at the system's components, the algorithm we developed to control them, and the challenges we faced during implementation.

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2. System Overview

The ICT3 system represents a generic automated industrial process designed to sort and assemble parts, perform inspections, and classify them as either accepted or rejected. The system consists of several areas that work in sequence to achieve these tasks: the sorting area, assembly chute, sensing area, and reject area. The system is controlled by a PC running LabView, which allows for real-time monitoring and troubleshooting of the automation process.

The primary functions of each system area are as follows:

- Sorting Area: Sorts metal pegs from plastic rings.
- Assembly Chute: Aligns plastic rings for assembly with metal pegs.
- Sensing Area: Inspects the assembly for accuracy.
- Reject Area: Accepts or rejects the assembly based on inspection results.

Sensors in each area provide feedback to the system, and actuators carry out physical tasks like sorting or rejecting parts. The entire process is controlled via LabView, with a user-friendly interface for system monitoring.

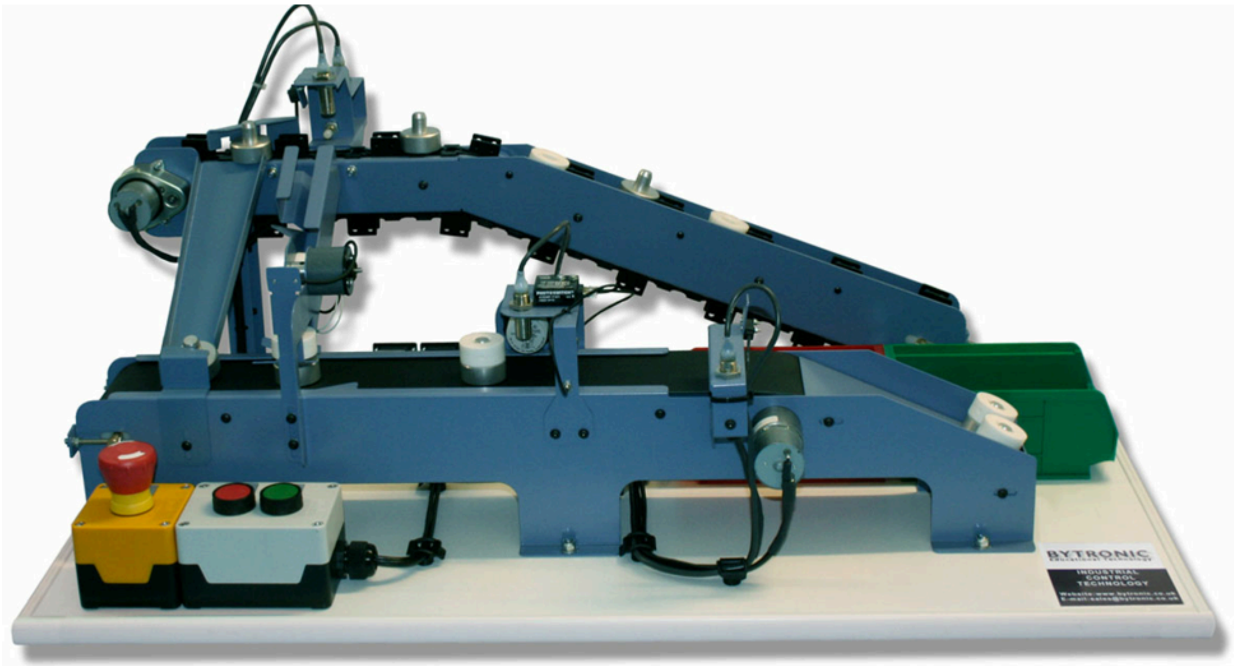

Figure 1: ICT3 System

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3. Sorting Area

Figure 2: Sorting Area Snapshot

The sorting area is the first step in the automated process. Here, parts are sorted as either metal pegs or plastic rings. The system uses a combination of sensors and actuators to perform this task:

1. Inductive Proximity Sensor: Detects metal objects, triggering the conveyor belt to move metal pegs to their designated path.
2. Infra-red Sensor: Detects plastic rings and activates a solenoid to push the rings into the assembly chute.
3. Sorting Solenoid: A mechanical actuator that directs the plastic rings to the assembly chute when detected by the infra-red sensor.

This process ensures that metal pegs and plastic rings are separated efficiently before moving onto the next stage.

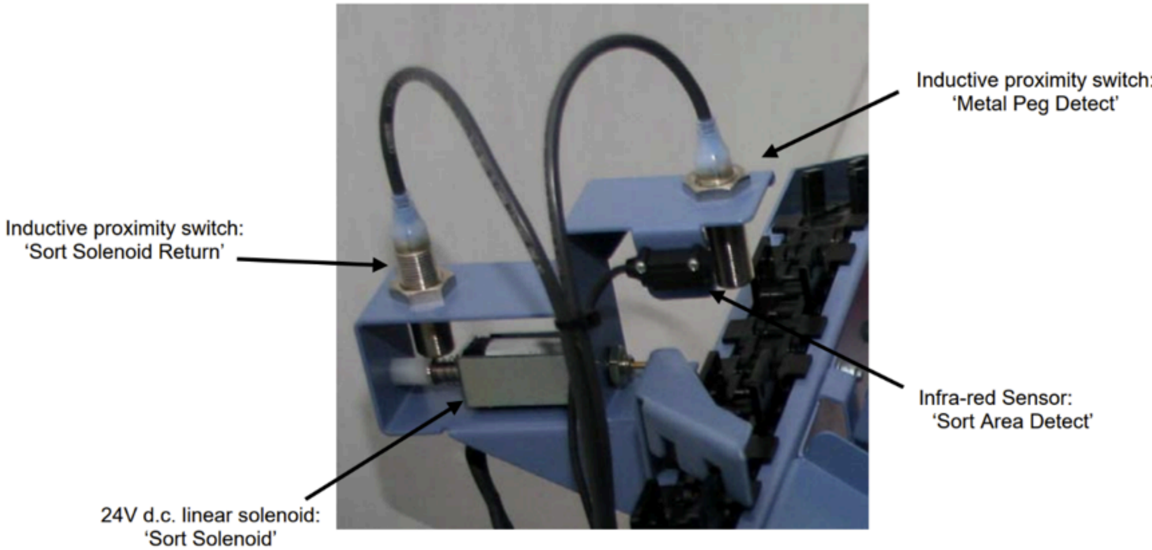

Figure 2: Sorting area

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4. Assembly Chute

Figure 3: Assembly Chute Snapshot

Sorted plastic rings move into the assembly chute, which buffers and aligns them for integration with metal pegs. A proximity sensor counts rings, ensuring only 5 rings are queued at a time. If the chute is full, excess rings bypass the chute. An infrared sensor triggers the chute's rotary solenoid, releasing rings for precise peg assembly. The assembly chute has several components:

1. Rotary Queue: Holds up to five rings in a buffer to prevent congestion.
2. Proximity Sensors: Count the rings in the queue and monitor the flow of the process.
3. Assembly Hopper: Aligns the rings so that they are properly positioned to be assembled with the metal pegs.

As each ring is dispensed, it is positioned correctly in the hopper for smooth engagement with the metal peg on the conveyor belt. This alignment ensures that assembly occurs without errors.

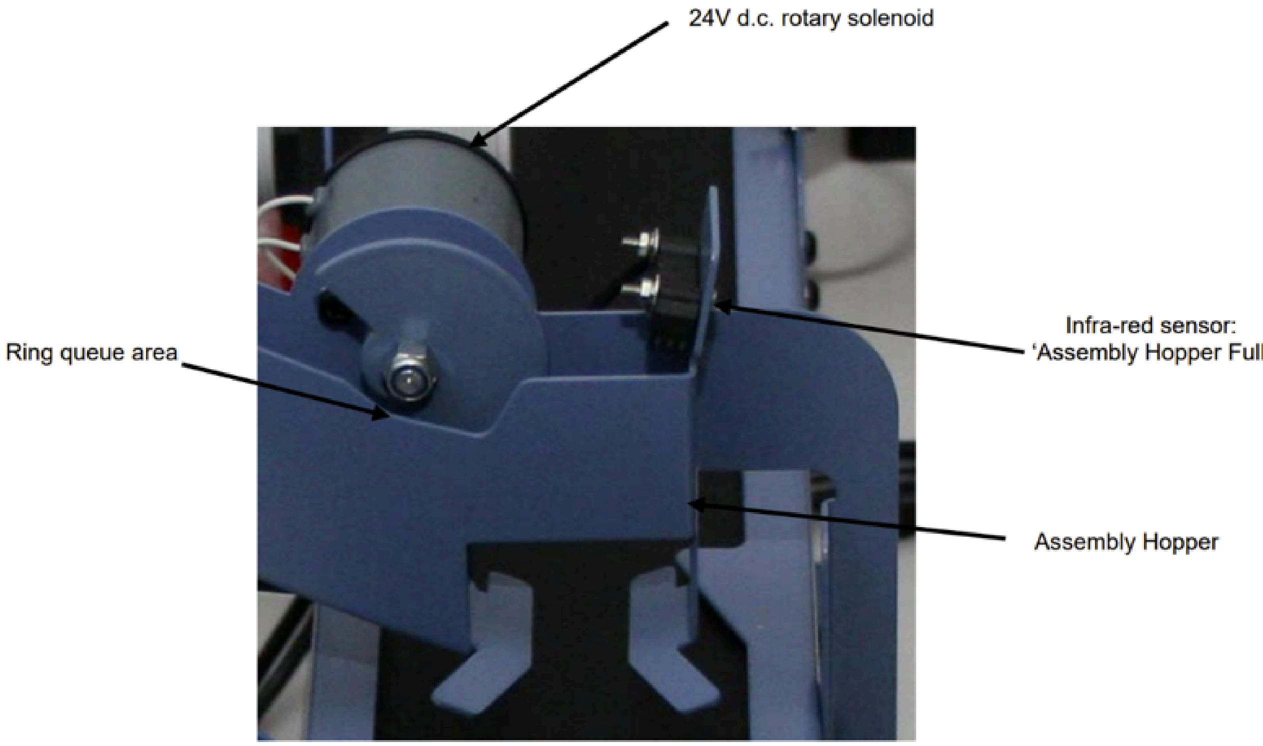

Figure 3: Assembly Chute

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5. Sensing Area

Figure 4: Sensing Area Snapshot

The sensing area is where the system verifies whether the assembly has been correctly completed. It uses three key sensors to perform this check:

1. Inductive Proximity Switch: Detects metal parts (pegs).
2. Capacitive Proximity Switch: Detects whether the part is correctly assembled.
3. Infra-red Fibre-Optic Sensor: Detects any part passing through, regardless of whether it is a peg or a ring.

The sensing area verifies assembly correctness using an inductive proximity switch, a capacitive sensor, and an infrared fiber-optic sensor. Three possible scenarios arise:

1. Correct Assembly: All sensors triggered.
2. Metal Peg Only: Inductive and infrared sensors triggered.
3. Plastic Ring Only: Infrared sensor triggered.

Data from this area informs the next stage, ensuring accurate decision-making in the reject process.

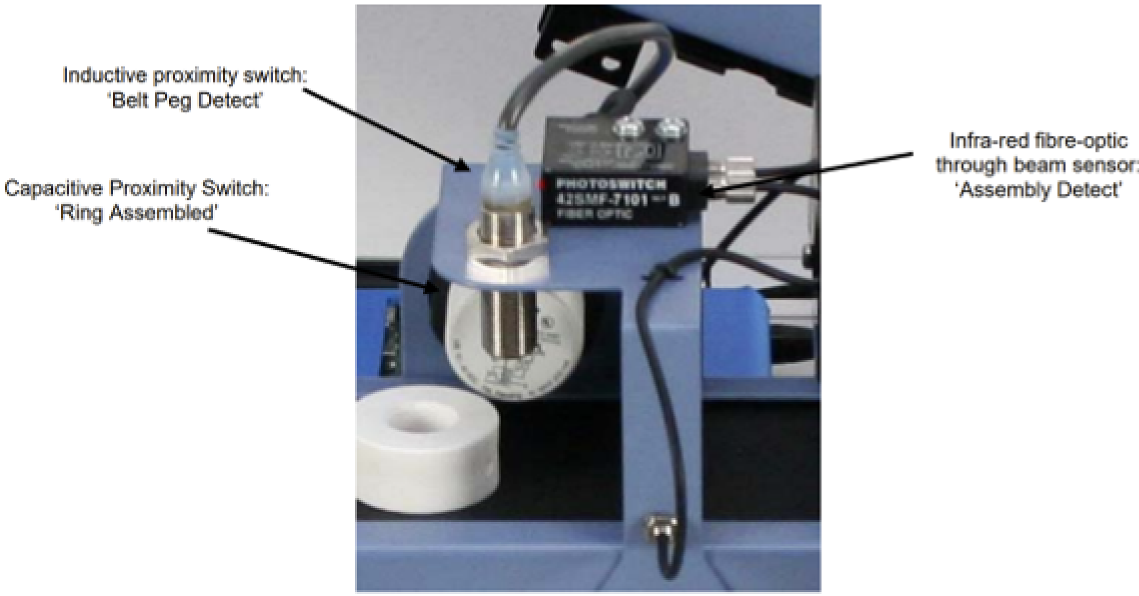

Figure 4: Sensing Area

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6. Reject Area

Figure 5: Reject Area Snapshot

The reject area is the final step of the process. Based on the feedback from the sensing area, parts are either accepted or rejected. This area consists of:

1. Infra-red Sensor: Detects the presence of a part.
2. Inductive Proximity Sensor: Monitors the solenoid's position to ensure correct operation.

In the reject area, assemblies are either accepted or rejected based on sensing data. An infrared sensor detects each item, while a solenoid either redirects faulty assemblies into a reject bin or allows correct assemblies to pass. The system tracks performance metrics, recording assembled and rejected items for efficiency calculations.

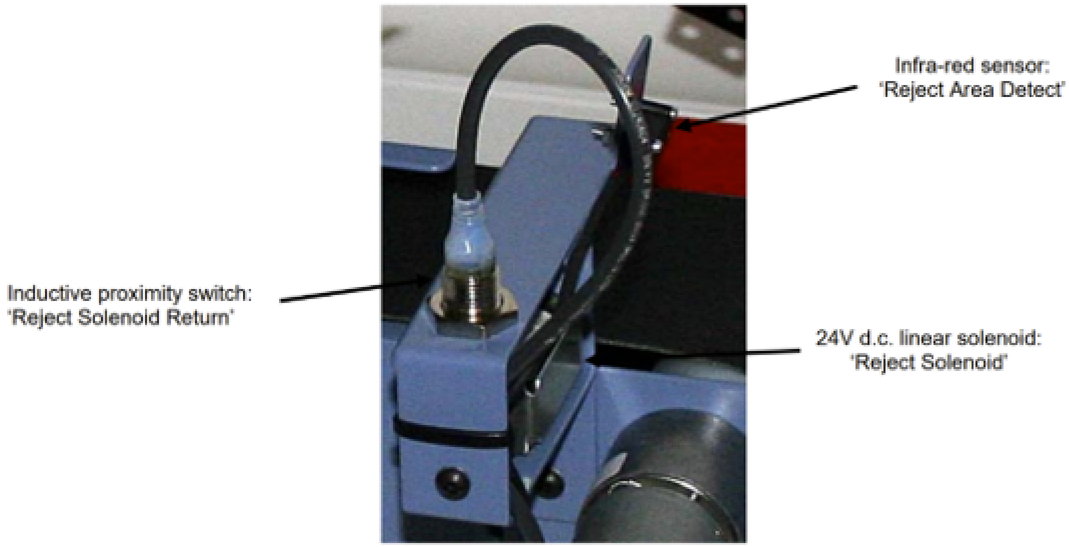

Figure 5: Reject Area

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7. Algorithms and Flowcharts

To automate the sorting, assembly, and inspection process, we developed a set of algorithms using LabView. These algorithms are visualized in the form of flowcharts, which guide the system's operations in each area.

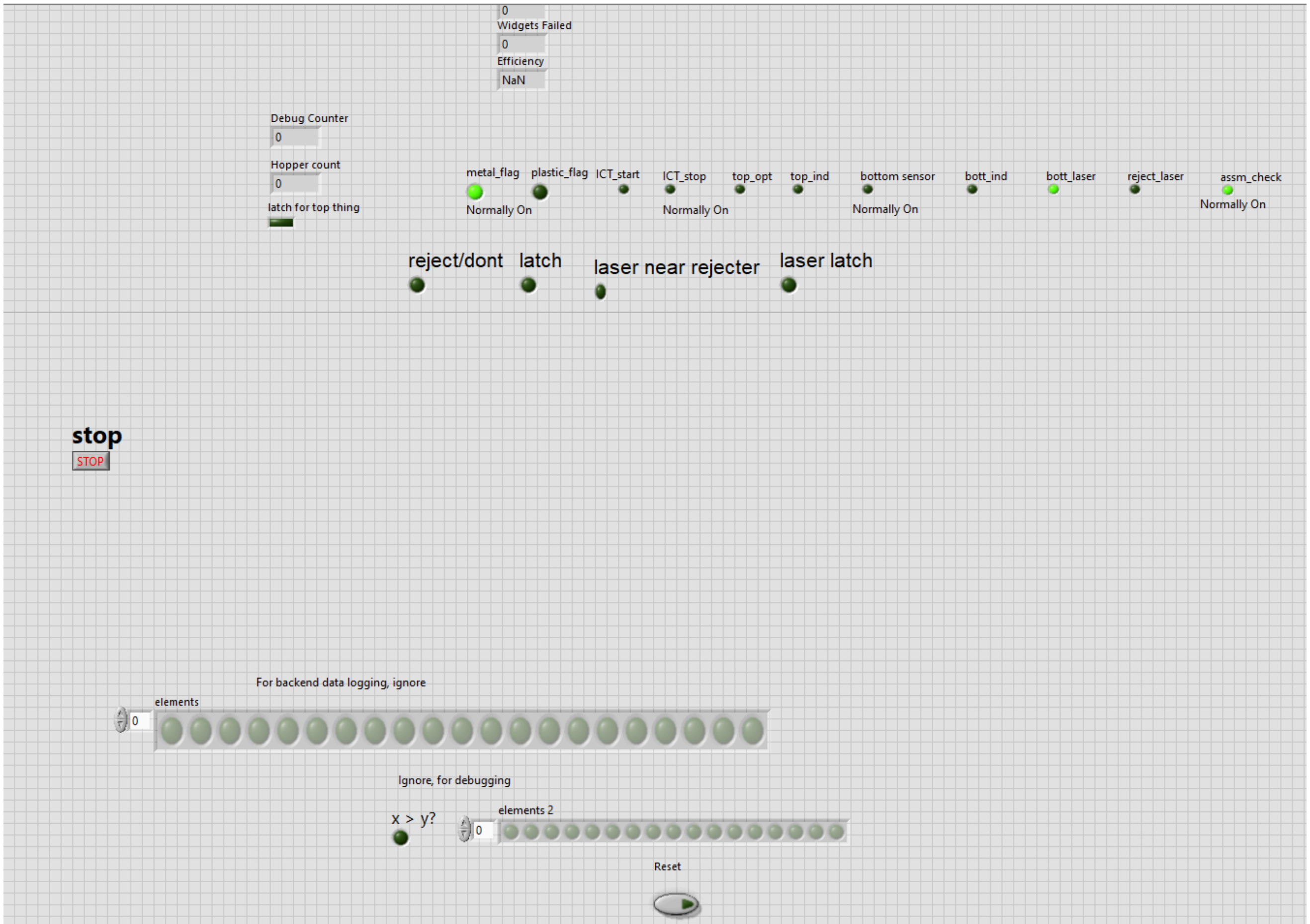

Figure 6: LabView Front Panel

The front panel of the LabView interface displays real-time information about the system's status. It shows the activation of sensors and actuators, the number of widgets processed, and the system's efficiency. Buttons for starting, stopping, and resetting the system allow the operator to control the process easily.

Key Flowcharts:

- Start Button Logic (Figure 7): Initializes the system and optionally uses data from previous runs.

Figure 7

- Stop Button Logic (Figure 8): Pauses the system to stop all conveyor belts and actuators.

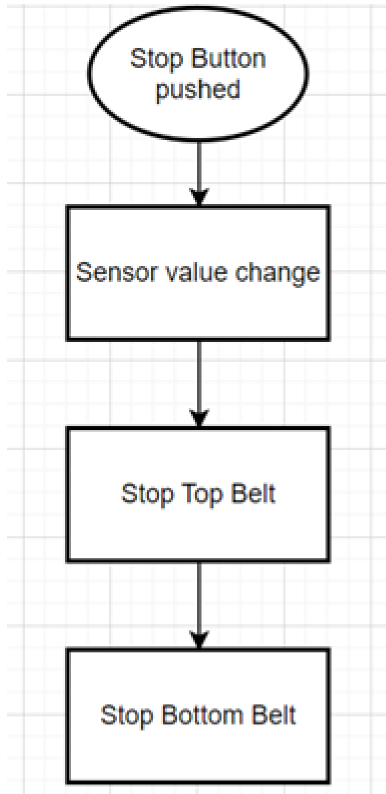

Figure 8

- Sorting and Assembly Logic (Figure 9): Determines how metal pegs and plastic rings are sorted and aligned in the assembly chute.

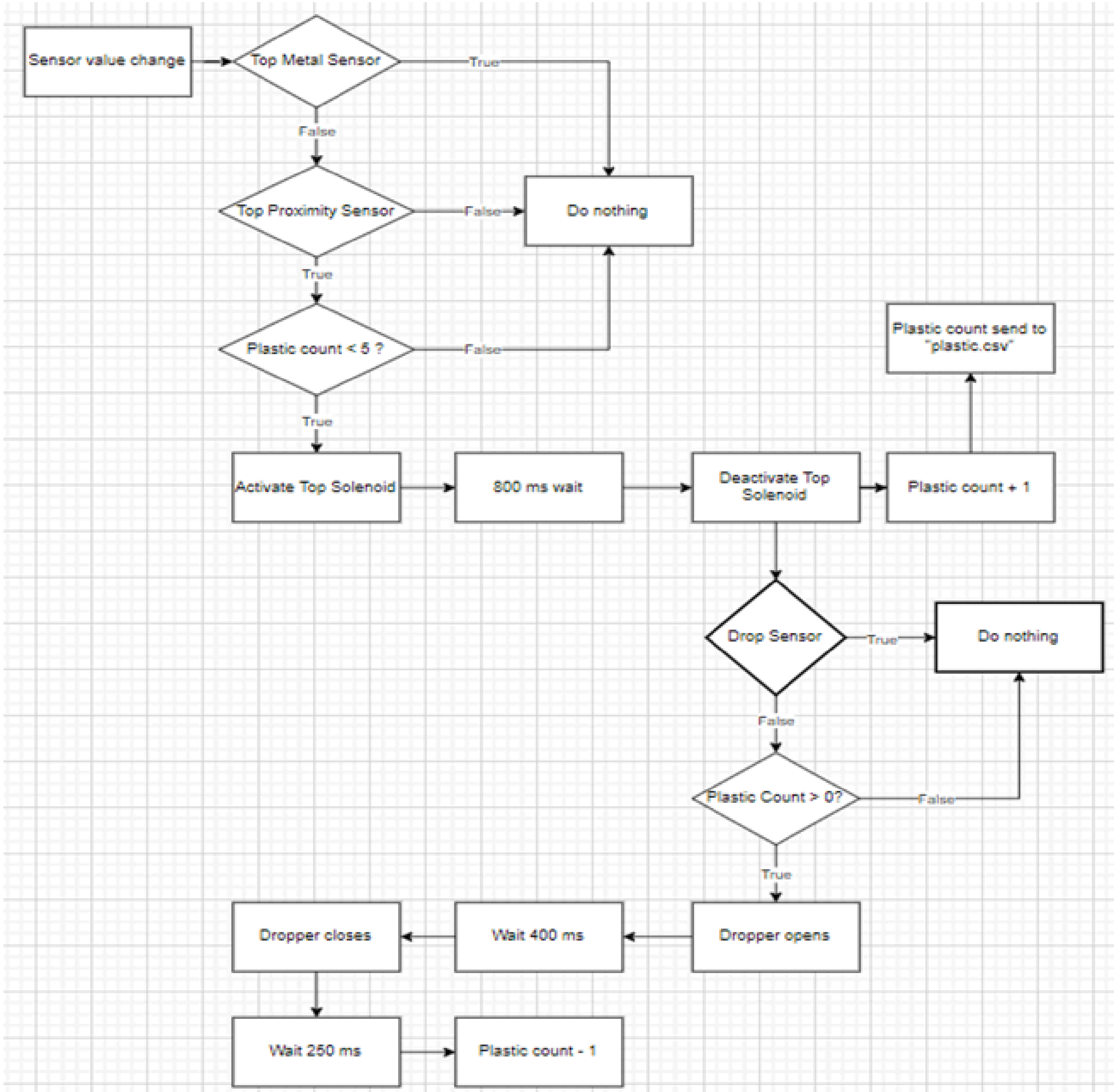

Figure 9

- Sensing Area Logic (Figure 10 & 11): Tracks the total number of widgets and distinguishes between properly assembled and improperly assembled parts.

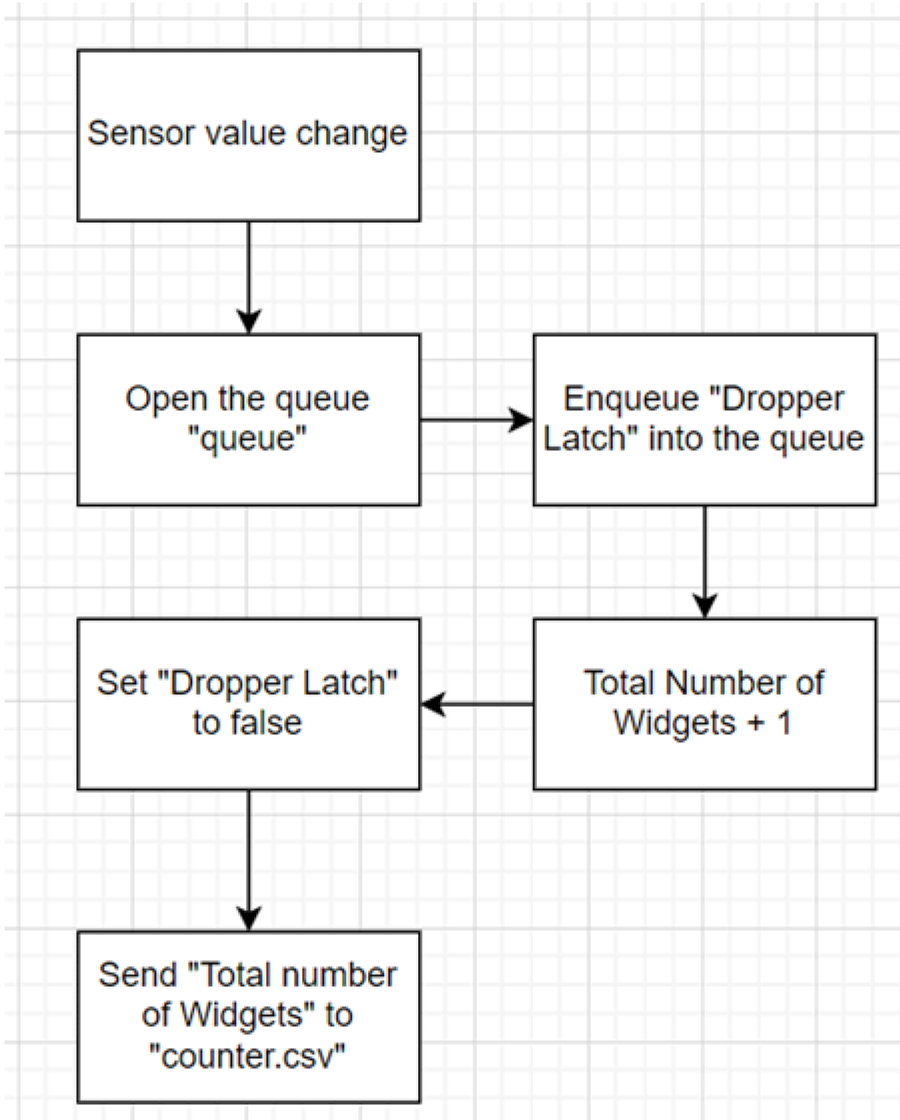
Sensor value change
Open the queue "queue"
Enqueue "Dropper Latch" into the queue
Total Number of Widgets + 1
Set "Dropper Latch" to false
Send "Total number of Widgets" to "counter.csv"

Figure 10

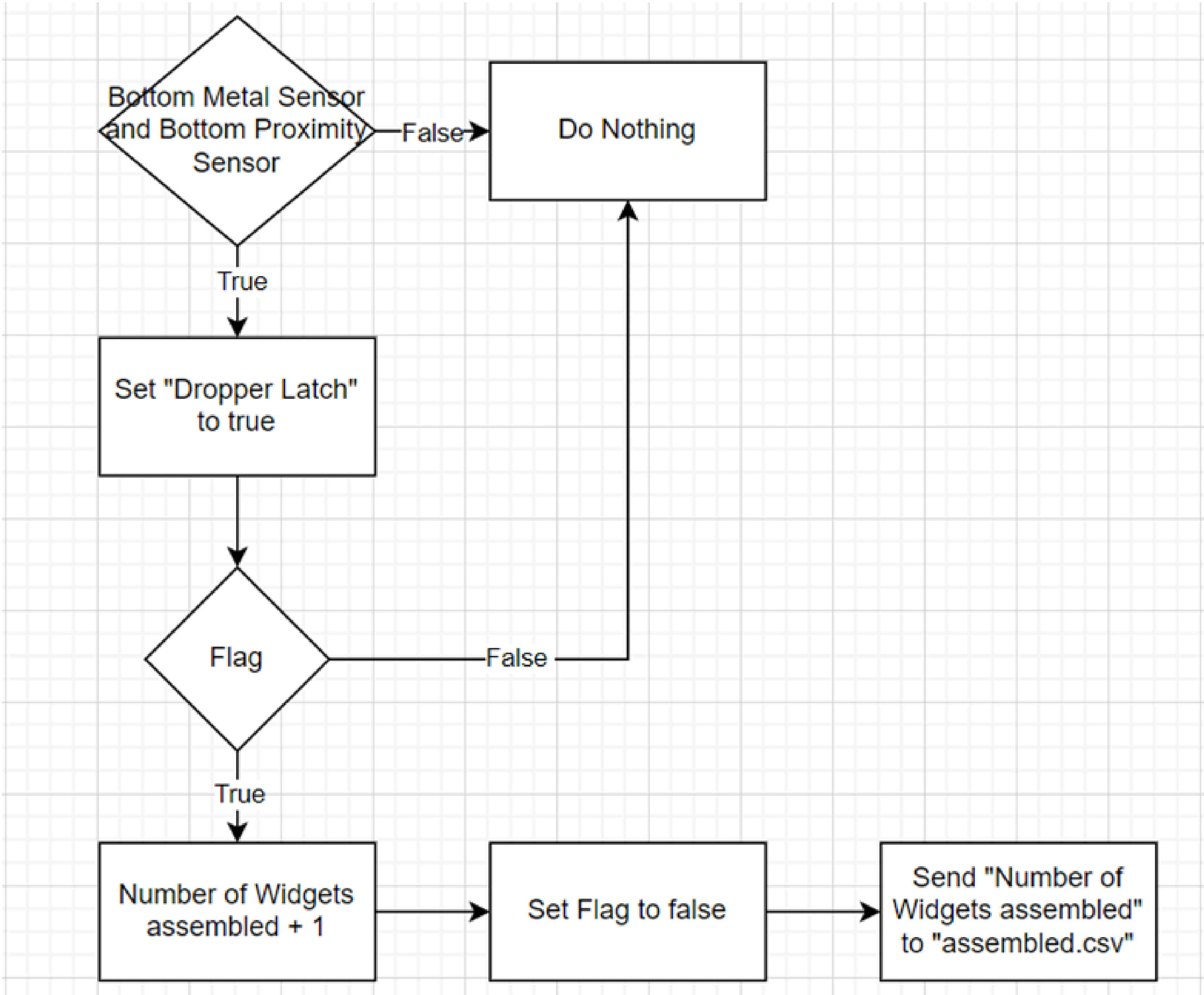

Figure 11

- Reject Area Logic (Figure 12): Ensures the correct parts are either accepted or rejected based on assembly quality.

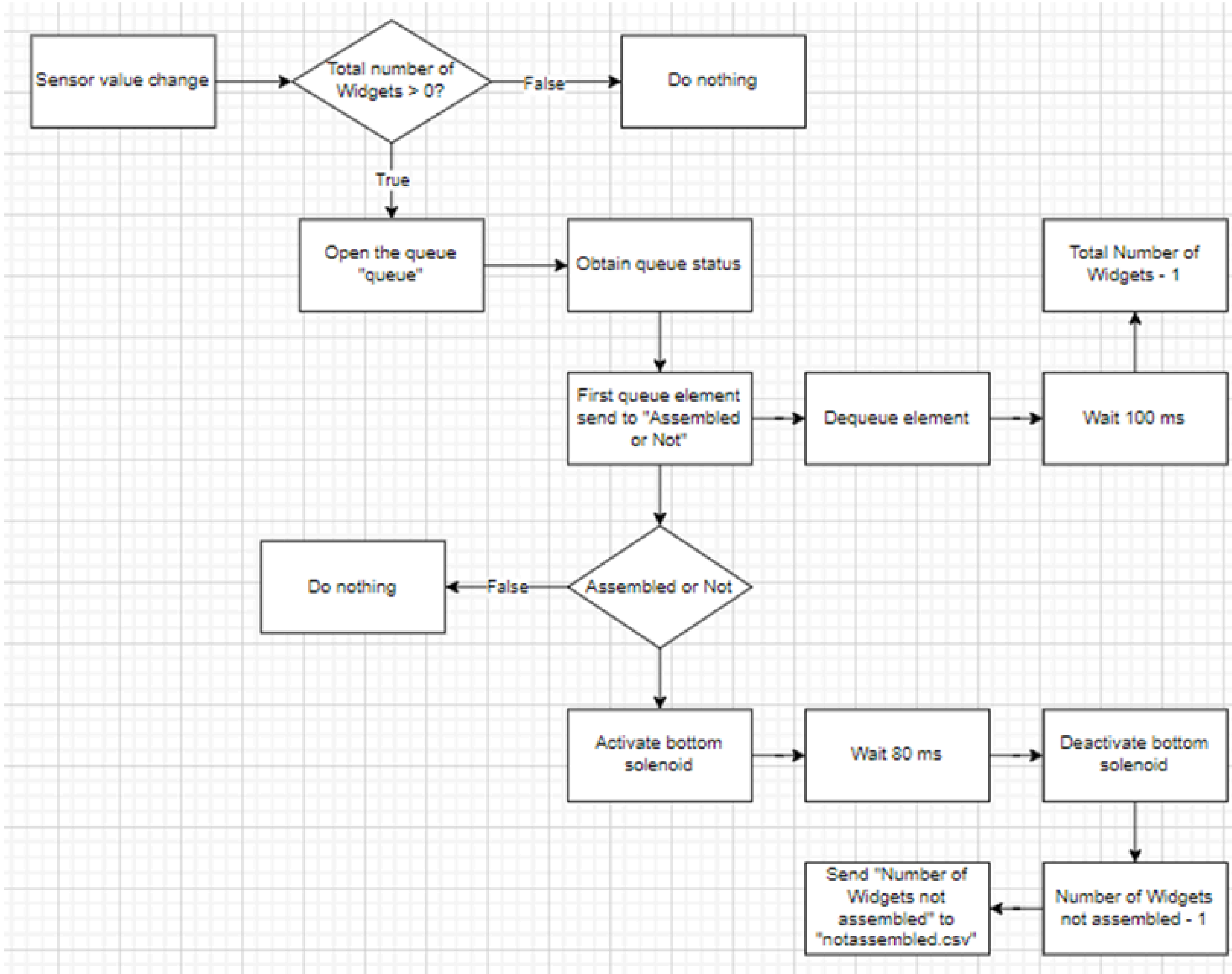

Figure 12

These flowcharts form the backbone of the system's algorithm, ensuring that each area of the process runs smoothly and efficiently.

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8. Challenges and Future Improvements

1. **Sensor Synchronization Issue**
   - Identified timing conflicts between Sorting Area and Sensing Area sensors
   - Proposed solution: Implement a priority event case for simultaneous sensor activations
2. **Hardware Limitations**
   - Experienced occasional sensor-to-software communication failures
   - Recommended thorough hardware diagnostics and potential sensor recalibration

3. **Potential Enhancements**
    - Develop more sophisticated error handling
    - Implement advanced machine learning algorithms for predictive maintenance
    - Improve real-time diagnostics and reporting

The project provided valuable insights into industrial automation, demonstrating the complex interplay between mechanical systems, sensors, and software control. While challenges were encountered, the AWA project successfully showcased the potential of integrated technological solutions in modern manufacturing.

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9. Conclusion

The ICT3 project provided valuable insights into industrial automation and the use of LabView for system control. While we faced challenges—including sensor conflicts, hardware limitations, and software learning curves—we successfully developed a system that automates the sorting, assembly, and inspection of parts. The experience not only improved our technical skills in algorithm design and system integration but also highlighted areas for future improvement, ensuring that the system can be made more reliable and efficient.